

Workshop - Week 2

Lecture 4

(a) Find the domain and range of the following functions:

(a) $f(x) = \sqrt{3-x} - \sqrt{2+x}$

(b) $g(x) = \sqrt{2-\sqrt{x}}$

(c) $h(x) = \frac{2x^3-5}{x^2+x-6}$

(b) Sketch the graph of the following functions:

(a) $f(x) = x + |x|$

(b) $g(x) = \begin{cases} |x| & \text{if } x \leq 1 \\ 1 & \text{if } x > 1 \end{cases}$

(c) $h(x) = ||x| - 1|$

(c) Determine whether f is even, odd, or neither:

(a) $f(x) = x^3 - x$

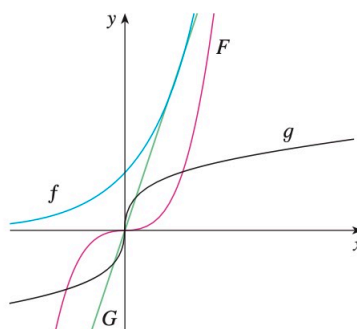
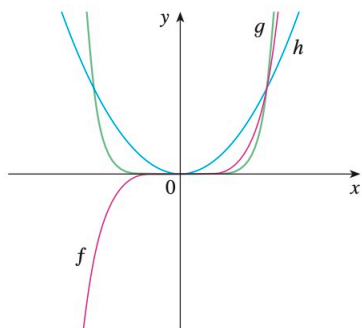
(b) $g(x) = \frac{1}{x^2+1}$

(d) If f and g are both even functions, what can you say about $f + g$ and $f \cdot g$? What if f is even and g is odd?

Lecture 5

(a) Match each equation with its graph, and explain your choice

(a) $y = x^2$ (b) $y = x^5$ (c) $y = x^8$ (d) $y = \sqrt[3]{x}$



(b) Find an expression for a cubic function f if $f(1) = 6$ and $f(-1) = f(0) = f(2) = 0$.

(c) Find $f \circ g \circ h$ of the following functions:

(a) $f(x) = x^2$, $g(x) = x^2 + 1$, $h(x) = x - 1$

(b) $f(x) = 3x - 2$, $g(x) = \sin x$, $h(x) = x^2$

(d) Suppose that g is an even function and let $h = f \circ g$. Is h even, odd, or neither?

(e) Suppose that g is an odd function and let $h = f \circ g$. Is h even, odd, or neither?

(f) Starting with the function $f(x) = e^x$, write down the equation of the graph that result from:

(a) reflecting the graph of f about the line $y = 3$,

(b) reflecting the graph of f about the line $x = 1$.

(g) Find the domain of the function $f(x) = \frac{1-e^{x^2}}{1-e^{1-x}}$.

(h) Find the domain of $f(x) = \ln(e^x - 4)$. Compute f^{-1} and find its domain.

(i) Find the domain and range of the function

$$g(x) = \arcsin(3x + 1).$$